

Summary for 2024-09-12-Duality

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1 Summary

This Linear Programming lecture focused on duality theory, exploring the relationship between primal and dual problems and their implications for optimization.

Primal Problem: A linear programming problem formulated as a maximization problem, aiming to maximize a linear objective function subject to linear inequality constraints and non-negativity constraints on the variables.

Dual Problem: A linear programming problem derived from the primal problem, formulated as a minimization problem with an objective function and constraints related to the primal problem's structure through matrix transposition and sign changes. Multiple equivalent formulations of the dual problem exist.

Weak Duality Theorem: The objective function value of any feasible solution to the primal problem is less than or equal to the objective function value of any feasible solution to the dual problem.

Matrix Representations of Primal and Dual Problems: Both primal and dual problems can be efficiently represented using matrices, highlighting the inherent relationship between their structures, specifically the negative transpose relationship between their constraint matrices.

Complementary Slackness Theorem: Primal and dual feasible solutions are optimal if and only if the product of each primal variable and its corresponding dual slack variable, and each dual variable and its corresponding primal slack variable, equals zero. This implies that at least one variable in each pair must be zero at optimality.

Strong Duality Theorem: If a primal linear programming problem has an optimal solution, then its dual problem also has an optimal solution, and their optimal objective function values are equal (zero duality gap). The proof provides a method for deriving the optimal dual solution from the optimal primal solution.

Four Possible Solution Outcomes: Four mutually exclusive scenarios exist for primal-dual problem pairs: both have optimal solutions with equal objective values; the primal is feasible and unbounded, while the dual is infeasible; the dual is feasible and unbounded, while the primal is infeasible; and both are infeasible.